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Data-Driven Methods for Multiple and Dual Response Optimization on Operational Data Using Stochastic PRIM

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ABSTRACT

Upon establishing multiple response optimization (MRO) or dual response optimization (DRO), response surface methodology (RSM) acts as a conventional model-driven framework for optimizing small-sized experimental data to derive an input condition in which a decent response is expected. RSM conducts multiple experiments to estimate response surfaces via empirical prediction models, but it suffers from predictive uncertainty upon fitting strict models onto large, complex operational data. Data-driven approaches on MRO and DRO have gained attention by employing supervised data mining algorithms to discover outstanding input subrange options from operational data, while effectively circumventing the predictive capability issues of RSM. In this study, we propose two new stochastic data-driven methods for response optimization to automatically derive numerous gratifying input conditions from industrial operational data. The random subspace method is incorporated into the patient rule induction method in accordance with valid objective function designs for MRO and DRO. Comparative case studies demonstrate the superior performance of our stochastic methods over deterministic baselines through N-fold validation and sensitivity analyses.

1 | Introduction

Response surface methodology (RSM) utilizes empirical models to identify an optimal condition from experimental data by building a functional, predictive relationship between input features and response values [1]. Multiple response surface optimization (MRSO) is a conventional RSM framework for multiple response optimization (MRO), where small-sized experimental data are repeatedly collected along with predesigned experiments to optimize several response variables of interest via linear regression models and steepest ascent [2]. Similarly, dual response surface optimization (DRSO) establishes dual response optimization (DRO) on experimental data by fitting parametric models to

derive an input condition where the response variable is dually optimized with respect to mean location and dispersion [3].

Owing to the widespread installation of on-site sensors in manufacturing processes, manufacturing process stages are commonly optimized nowadays by automatically collected operational datasets that are huge in volume, instead of using experimental data [4]. Yet, it is notable that predicting the true global underlying structure of large-scale tabular data is challenging in general [5]. Operational data is usually characterized by many observational rows and several column features, implying a complex underlying structure (i.e., multicollinearity, high input dimensionality, multiple global maxima, measurement errors,

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TABLE 1 | Transformation of a realized response value $y_{i,j}$ into a desirability value $d_{i,j}$ with respect to the range of $y_{i,j}$ and its inherent characteristics (LTB/STB/NTB).

Y_j	Y_j : Larger-The-Better (LTB)			Y_j : Smaller-The-Better (STB)			Y_j : Nominal-The-Better (NTB)		
Range of $y_{i,j}$	$y_{i,j} \leq T_j^{lower}$	$T_j^{lower} < y_{i,j} < T_j^{upper}$	$y_{i,j} \geq T_j^{upper}$	$y_{i,j} \leq T_j^{lower}$	$T_j^{lower} < y_{i,j} < T_j^{upper}$	$y_{i,j} \geq T_j^{upper}$	$y_{i,j} \leq T_j^{lower}$ or $y_{i,j} \geq T_j^{upper}$	$T_j^{lower} < y_{i,j} < T_j^{Target}$	$T_j^{Target} \leq y_{i,j} < y_j^{upper}$
Value of $d_{i,j}$	0	$\left(\frac{y_{i,j} - T_j^{lower}}{T_j^{upper} - T_j^{lower}} \right)^{s_j}$	1	1	$\left(\frac{T_j^{upper} - y_{i,j}}{T_j^{upper} - T_j^{lower}} \right)^{s_j}$	0	0	$\left(\frac{y_{i,j} - T_j^{lower}}{T_j^{Target} - T_j^{lower}} \right)^{s_j}$	$\left(\frac{T_j^{upper} - y_{i,j}}{T_j^{upper} - T_j^{Target}} \right)^{t_j}$

etc.) which makes it difficult for a parametric RSM to approximate the response surfaces. Notably, MRSO and DRSO often fail when applied to operational data for surface predictions due to the low performance of linear regression models [3, 6].

Unlike such model-driven RSM specialized for optimizing small-sized experimental data, data-driven approaches employ data mining algorithms for response optimization by flexibly identifying local input conditions within large operational data. Related literature shows that algorithmic features of the classification and regression tree (CART) or the patient rule induction method (PRIM) effectively optimize operational data by searching for gratifying input conditions in a data-driven way, while without a need to construct empirical prediction models [6–10]. In this paper, we propose two new data-driven methods for optimizing large-scale manufacturing data by hybridizing stochasticity of the random subspace method and the PRIM algorithm in accordance with suitable objective function designs for MRO and DRO. Section 2 provides a literature review on data-driven response optimization methods along with several objective functions as stepstones for Section 3. Section 3 then introduces two new stochastic data-driven methods for MRO and DRO. A comprehensive case study is conducted in Section 4 to validate the performance of the two methods in comparison with other baseline methods via N-fold cross-validation and sensitivity analyses. Section 5 concludes the paper by discussing future research topics within the field of data-driven response optimization.

2 | Literature Review

2.1 | Review of Objective Function Designs for Data-Driven MRO and DRO

In Section 2.1, the desirability metric and mean squared error (MSE) are reviewed as representative designs on objective functions for MRO and DRO, respectively. Harrington [11] proposed the fundamental idea of the desirability function approach, which was further developed by Derringer and Suich [12]. Consider a vector of multiple response variables $Y = \{Y_1, \dots, Y_j, \dots, Y_J\}$. The realized values of the i th observation ($i : 1 \sim N$) can then be denoted by $y_i = \{y_{i,1}, \dots, y_{i,j}, \dots, y_{i,J}\}$, where $y_{i,j}$ is the realized value of the j th response Y_j ($j : 1 \sim J$). Table 1 summarizes the desirability function designs for a larger-the-better (LTB), smaller-the-better (STB) and nominal-the-better (NTB) response variable, respectively. The desirability function design for Y_j is determined by several parameters: $T_j^{lower}/T_j^{upper} / T_j^{Target}$

(target parameter), w_j (weight parameter) and s_j / t_j (shape parameter). Notably, the exact values of the parameters are assumed to be known a priori by predefined product standards, specification limits, or relevant domain knowledge of the process engineer [13].

By denoting a vector $d_i = \{d_{i,1}, \dots, d_{i,j}, \dots, d_{i,J}\}$ as J desirability values of the i th observation, d_i can be aggregated into a scalar desirability D_i using the overall desirability function in Equation (1) as a weighted geometric mean. By applying the overall desirability function across multiple desirability values, a multi-objective optimization problem is eventually simplified into a single-objective optimization problem [2]. The objective function for an input subrange (or subregion) can then be denoted as $\bar{D}$, computed as an average value of D_i among observations existing within that subrange. A major advantage of employing desirability functions for objective function design in MRO is that multivariate response variables with differing scales can eventually be transformed into a unified scale while considering their inherent characteristics. The desirability function approach is thus considered highly suitable for MRO [4, 14–16].

$$D_i = f(d_{i,1}, \dots, d_{i,J}, w_1, \dots, w_J) = \left(\prod_{j=1}^J d_{i,j}^{w_j} \right)^{\frac{1}{\sum_{j=1}^J w_j}} \quad (i : 1 \sim N). \quad (1)$$

Meanwhile, DRO aims to optimize both the target location (mean) and dispersion scale (standard deviation) of the response values by identifying an input subrange in which the response variable is dual-optimized. Accordingly, the *MSE* metric is a suitable objective function design for DRO in that it models both location and scale of the response variable. Equation (2) denotes how the *MSE* objective value of a subregion or input subrange can be calculated with respect to a univariate response Y . A subregion *MSE* consists of two terms: i) a squared response bias $(\bar{y}_{Subregion} - Target)^2$ indicating the difference between $\bar{y}_{Subregion}$ (sample mean of y_i among the within-subregion observations) and a predefined target constant (*Target*), and ii) $\hat{\delta}_{Subregion}^2$ indicating sample variance of y_i values within a subregion. Unlike the desirability objective $\bar{D}$ which is maximized for data-driven MRO, *MSE* acts as a minimization objective for data-driven DRO. Notably, the objective functions $\bar{D}$ and *MSE* are computed only by using the observed response vector y_i , without any predicted $\hat{y}_i$ values. Therefore, such objective designs are highly suitable for data-driven response optimization in that prediction models such as linear regression are not required.

$$MSE_{Subregion} = (\bar{y}_{Subregion} - Target)^2 + \hat{\delta}_{Subregion}^2. \quad (2)$$

2.2 | Review of Data-Driven Methods for Response Optimization

In Section 2.2, some representative methods for data-driven response optimization are reviewed: multi-response PRIM (MR-PRIM, [17]), dual-response PRIM (DR-PRIM, [3]) and multi-response CART (MR-CART, [8]). The PRIM algorithm itself was introduced as a bump-hunting method by Friedman and Fisher [18] to identify the subspaces of the input domain in which the density or value of the output is significantly different from a typical extent. This aspect of PRIM has led to its modified usage for industrial process optimization to discover optimal input subspaces with desirable outputs. Accordingly, various PRIM-based methods including MR-PRIM [17], multistage PRIM [19], dual response PRIM (DR-PRIM, [3]), extended multi-response PRIM (EMR-PRIM, [10]), and exponentially weighted moving average PRIM (EWMA-PRIM, [6]) have been reported in the data-driven response optimization literature.

With respect to MR-PRIM and DR-PRIM, the former optimizes multiple responses with respect to desirability mean while the latter dual-optimizes a single response variable. MR-PRIM optimizes a tabular operational dataset consisting of N row observations, J multivariate responses $Y = \{Y_1, \dots, Y_J, \dots, Y_J\}$, and P input features $X = \{X_1, \dots, X_p, \dots, X_p\}$. An i th realized observation row can thus be denoted as $Obs_i = \{y_i, x_i\} = \{y_{i,1}, \dots, y_{i,J}, \dots, y_{i,J}, x_{i,1}, \dots, x_{i,p}, \dots, x_{i,p}\}$. Equation (1) is then applied onto the i th ($i: 1 \sim N$) realized response vector y_i , so that $Obs_i = \{D_i, x_i\} = \{D_i, x_{i,1}, \dots, x_{i,p}, \dots, x_{i,p}\}$ is assumed for MR-PRIM. For DR-PRIM, $Obs_i = \{y_i, x_i\} = \{y_i, x_{i,1}, \dots, x_{i,p}, \dots, x_{i,p}\}$ with a univariate response variable Y .

Prior to fitting the embedded PRIM, all rows of the dataset are randomly split into training and testing dataset. The fitting procedure for both methods then begins with an initial box denoted by B_0 , a P -dimensional hyperrectangle box including every observation in the training dataset. B_0 is then consecutively peeled by simply eliminating $100\alpha\%$ of observations with the lowest (or highest) percentile values of a particular input feature. α is a peeling hyperparameter predefined by the process engineer as a small value (e.g., 0.05) for patient peeling. After m peelings are performed, the input subrange of the current box B_m can be expressed as an axis-aligned hyperrectangle in Equation (3). $[\Delta_{m,p}] = [LB_{m,p}, UB_{m,p}]$ as in Equations (3,4) denotes the min-max subrange of X_p among observations within B_m .

$$B_m = [\Delta_{m,1}] \times [\Delta_{m,2}] \times \dots \times [\Delta_{m,p}] \times \dots \times [\Delta_{m,p}]. \quad (3)$$

$$LB_{m,p} = \min_i \{x_{i,p} | Obs_i \in B_m\}, UB_{m,p} = \max_i \{x_{i,p} | Obs_i \in B_m\}. \quad (4)$$

C_{m-1} in Equation (5) is a candidate set of $2P$ peeled boxes, from which the next peeled box B_m is determined. Equations (6) and (7) denote the input subranges of candidate boxes $C_{m-1}^{(p-)}$ and $C_{m-1}^{(p+)}$. $C_{m-1}^{(p-)}$ denotes the box state where observations of B_{m-1} with a lower $0\% \sim 100\alpha\%$ percentile value of X_p have been eliminated from B_{m-1} . Likewise, $C_{m-1}^{(p+)}$ denotes the same but with respect to upper $100(1-\alpha)\% \sim 100\%$ percentile values. $Pr_{100\alpha}$ in Equation (8) refers to the 100α th sample percentile value. Among $2P$ candidate boxes constituting C_{m-1} , the box with a highest $\bar{D}$

is determined as B_m . DR-PRIM does the same, but with respect to the lowest MSE . For both methods, each peeling step is a locally optimal rule-based decision without involving any strict model-based assumptions.

$$C_{m-1} = \{C_{m-1}^{(1-)}, C_{m-1}^{(1+)}, \dots, C_{m-1}^{(p-)}, C_{m-1}^{(p+)}, \dots, C_{m-1}^{(P-)}, C_{m-1}^{(P+)}\}. \quad (5)$$

$$C_{m-1}^{(p-)} = [\Delta_{m-1,1}] \times \dots \times [\Delta_{m-1,p-1}] \times [\Delta_{m-1,p}^{(-)}] \times [\Delta_{m-1,p+1}] \times \dots \times [\Delta_{m-1,P}]. \quad (6)$$

$$C_{m-1}^{(p+)} = [\Delta_{m-1,1}] \times \dots \times [\Delta_{m-1,p-1}] \times [\Delta_{m-1,p}^{(+)}] \times [\Delta_{m-1,p+1}] \times \dots \times [\Delta_{m-1,P}]. \quad (7)$$

$$\begin{aligned} [\Delta_{m-1,p}^{(-)}] &= [Pr_{100\alpha} \{x_{i,p} | Obs_i \in B_{m-1}\}, UB_{m-1,p}], [\Delta_{m-1,p}^{(+)}] \\ &= [LB_{m-1,p}, Pr_{100(1-\alpha)} \{Obs_i | x_{i,p} \in B_{m-1}\}]. \end{aligned} \quad (8)$$

If an m th peeling step is the first step to yield a box with a smaller support than a stopping hyperparameter β_0 , the peeling step stops and the box having a best box objective value within the box trajectory ($B_0 \rightarrow B_1 \rightarrow \dots \rightarrow B_{m-1}$) can be selected as an optimal subregion within that peeling trajectory. The trajectory is then regarded as a single PRIM-based trial. Notably, input conditions $B_0 \sim B_{m-1}$ are referred to as training set subregions since they have been discovered from the training set. A corresponding subregion within the testing set can then be uniquely defined by applying the min-max input subrange of a training set subregion to the entire testing set. Accordingly, box objective of a derived subregion for PRIM-based methods can be calculated for the training and testing set, respectively. If there is not a significant difference in the box objective value between the two, the subregion is validated and can thus be utilized for actual process control.

Lee and Kim [17], the authors of MR-PRIM, recommend selecting the validated box with the highest testing set $\bar{D}$. Yet, the training set subregion typically contains a far larger number of observations than the testing set subregion, making its $\bar{D}$ a more reliable indicator of desirability. $\bar{D}$ of the testing set subregion thus may serve only as an ancillary metric to assess whether a significant difference exists between the training and testing set subregion [9]. For DR-PRIM, the process engineer can freely discard a subregion by discretion if there seems to be a significant difference in the box MSE value between the training and testing subregion [3]. Yet, it is notable that a dual-validation is also feasible by applying t -test (for testing location) and F -test (for testing variance), or alternatives such as Mann-Whitney test and Levene's test.

The CART algorithm was introduced by Breiman et al. [20] as a binary decision tree framework. CART utilizes the fundamental idea of divide-and-conquer, recursively partitioning the entire dataset into numerous nodes containing similar response values. Building on this foundation, MR-CART [8] was proposed as a novel MRO method that leverages CART to identify desirable input conditions from operational data. An i th observation with P input features can be denoted as $Obs_i = \{D_i, x_i\} = \{D_i, x_{i,1}, \dots, x_{i,p}, \dots, x_{i,p}\}$. The root node acts as a parent node ($Node_{parent}$) and is split into two child nodes ($Node_{left}, Node_{right}$). The two child nodes then act again as parent

nodes and are recursively split into child nodes to eventually yield a partitioned tree. The optimal input feature (X_{p^*}) and its threshold value (θ^*) upon partitioning are determined by Equation (9), minimizing the within-node variability of desirability across two child nodes. The child nodes partitioned by X_p are $Node_{left}(p, \theta) = \{Obs_i | x_{i,p} \leq \theta, Obs_i \in Node_{parent}\}$ and $Node_{right}(p, \theta) = \{Obs_i | x_{i,p} > \theta, Obs_i \in Node_{parent}\}$. $\bar{D}_{left}$ and $\bar{D}_{right}$ denote the sample mean value of D_i for $Node_{left}$ and $Node_{right}$, respectively.

$$p^*, \theta^* = \underset{p, \theta}{\operatorname{argmin}} \left[\sum_{Obs_i \in Node_{left}(p, \theta)} \{D_i - \bar{D}_{left}\}^2 + \sum_{Obs_i \in Node_{right}(p, \theta)} \{D_i - \bar{D}_{right}\}^2 \right]. \quad (9)$$

When learning is completed under predefined restrictions (i.e., tree depth, minimum leaf node size) the nodes at the very end of the tree without having child nodes are referred to as terminal nodes. A statistical validation (e.g., a two-sample t -test) is then applied to every terminal node to verify whether a significant difference in desirability exists between the subregions from the training and testing set. Among the subregions that survive the validation by not being rejected, those with the highest desirability can be selected by the process engineer as optimal input conditions.

Questioning whether patient PRIM-based methods can outperform relatively aggressive CART-based methods upon discovering a better input condition or vice versa cannot yield a generalized answer, as they fundamentally differ in how the input space is searched [9]. Yet, they both successfully circumvent the predictive uncertainty issues of MRSO or DRSO by avoiding the construction of prediction models and computing the objective function using the observed Y_i values only. Meanwhile, the PRIM and CART based methods reviewed above are inherently deterministic in that only a single-fixed algorithmic trial can be implemented onto the given training dataset. Yet, a single trial may not ensure that it eventually derives or approximates the unknown global optimum (or optima) of a complex operational dataset. Therefore, it can be inferred that empowering the algorithmic effort for an intensive search beyond a single trial can make it more likely to discover better input conditions within identical data, which may have remained undiscovered via single deterministic trial.

2.3 | Review on Random Subspace Technique and Related Applications

Pioneering authors, including Amit and Geman [21] and Ho [22], suggested the fundamental idea of applying the random subspace technique to advanced data mining. Suppose that a P number of input features $X = \{X_1, \dots, X_p, \dots, X_P\}$ exists; the random subspace of X can be defined as $R(< P)$ subsampled number of input features among X . Breiman's [23] random forest algorithm is a prediction machine that utilizes the random subspace method along with non-parametric bootstraps as a means of enhancing the predictive performance of decision trees. By allowing each of the multiple decision trees to be learned

using a random subset of input features at every node split, the trees become less correlated with each other upon learning the bootstrapped data. The predictions of the trees are then aggregated, eventually reducing the model variance to ensure better predictive performance than a single CART tree [24].

Notably, scenario discovery is a field of study in which the random subspace method is frequently integrated with PRIM to enhance the interpretability of input spaces. The outcome is typically defined as a binary indicator (of interest/not of interest), and the goal is to identify input spaces that contain outcomes with a high purity of interest in a high-dimensional uncertainty space [25]. Accordingly, PRIM is often utilized to identify simple and interpretable boxes in an input space where the outcomes of interest are densely concentrated. For instance, Kwakkel and Cunningham [26] modified PRIM for scenario discovery by randomly sampling a subset of input features once per single PRIM trial. By repeating these randomized PRIM trials, boxes that ensure decent interpretability can be selected as optimal. Nassih and Berrado [27] introduced a binary classification method that blends PRIM with a random subspace technique to enable the interpretable discovery of subgroups.

This stochastic aspect of the random subspace method can in turn be utilized to optimize operational data by diversifying the peeling paths of PRIM. Suppose that a deterministic PRIM (i.e., MR-PRIM, DR-PRIM) has discovered a decent input condition for process control, while other better options are unidentifiable. If another PRIM-based method identifies validated input conditions with better box objective values than deterministic PRIM from identical operational data, it disproves the assumption that a single-fixed PRIM trial has invested a sufficient computational effort upon approximating the unknown global optimum (or optima). Accordingly, it can be inferred that adding stochasticity upon each peeling path of MR-PRIM or DR-PRIM can help diversify the search trajectories and increase the likelihood of discovering superior subregions than would be found through a single non-stochastic run. Based on this rationale, the following section introduces new data-driven stochastic PRIMs for MRO and DRO.

3 | Proposed Methods: MRS-PRIM and DRS-PRIM

Multiple response stochastic PRIM (MRS-PRIM) is a data-driven MRO method specialized for discovering outstanding input conditions from large operational data, by incorporating the stochasticity of the random subspace technique into the peeling steps of MR-PRIM via desirability function approach. Likewise, dual response stochastic PRIM (DRS-PRIM) is a stochastic extension of DR-PRIM for discovering dually optimal subregions from operational data. Process engineers can implement process control by selecting among the multiple gratifying subregion options identified by MRS-PRIM or DRS-PRIM, and can then adjust the subrange of controllable input features to those options. To the best of our knowledge, no PRIM-based methods have been proposed in the MRO literature for optimizing operational data via hybridizing input feature stochasticity with PRIM under an aggregated desirability metric, nor in the DRO literature with respect to stochastic PRIM under the MSE box objective. Note that the terms input subrange, input condition, subregion,

hyperrectangle box and terminal node are used interchangeably throughout the remainder of this paper, all referring to the subrange information of multiple input features.

Step 1. Preparing and preprocessing an operational dataset for response optimization

In Step 1, a tabular operational dataset is prepared for response optimization. The tabular notations for MRS-PRIM follow that of MR-PRIM, denoted by N row observations, J response variables, and P input features. The i th ($i : 1 \sim N$) row is thus denoted as $Obs_i = \{D_i, x_i\} = \{D_i, x_{i,1}, \dots, x_{i,p}, \dots, x_{i,p}\}$ after applying Equation (1). Likewise, the tabular notation is shared in between DR-PRIM and DRS-PRIM so that $Obs_i = \{y_i, x_i\} = \{y_i, x_{i,1}, \dots, x_{i,p}, \dots, x_{i,p}\}$ with a univariate response variable Y . All the rows of the operational dataset are randomly split into two subsets, and the split ratio is typically set in a way that the row size of the training set (used for discovering identified subregions) is larger than that of the testing set (used for confirmation) [16].

Notably, defining a subregion identified from the training set as a “training set subregion” and the corresponding input subrange in the testing set as a “testing set subregion” may be potentially misleading within the context of data-driven MRO or DRO. Such terminology may cause readers to misconceive a data-driven response optimization task as a “predictive task”. Assuming that the given operational dataset is large enough to represent the true unknown population in general, its numerous observations can be regarded as near-population data rather than small experimental samples of RSM. Accordingly, the aim of data-driven response optimization is not to “predict” unseen test data, but only to “identify” or “discover” subregions from the given operational data where the observed response values y_i are highly satisfactory. Minimizing the test prediction errors is not a valid objective of data-driven MRO or DRO, as the objective function value is computed directly from the observed response values, not from predicted values. Henceforth, we redefine the terms “training set subregion” and “testing set subregion” as “identified subregion” and “confirmation subregion”. The “identified subregion” comprises the observed response values Y_i used to compute the box objective in the optimization stage, while the “confirmation subregion” serves solely as an ancillary metric to validate whether the box objective of the “identified subregion” is robust to data splitting.

Step 2. Defining α (peeling hyperparameter) and β_0 (stopping hyperparameter)

In Step 2, hyperparameter values α and β_0 are predefined by the process engineer before fitting MRS-PRIM or DRS-PRIM to the operational dataset. As explained in Section 2.2, α is a peeling hyperparameter that determines the very lower ($0\% \sim 100\alpha\%$) or upper ($100(1 - \alpha)\% \sim 100\%$) percentiles of a certain input feature to be eliminated upon every rule-based peeling step. This hyperparameter is typically set as equal to or below 0.1 for patient peeling. β_0 is a stopping hyperparameter, defined as a ratio of support size between the current box

and the initial box. In general, a lower β_0 value raises the possibility of discovering narrower subregions with a better response objective value but also increases the risk of failure upon subregion validation [9]. This in turn implies that if a method consistently finds many validated subregions regardless of how small β_0 is, the method proves its robustness to subregion validation.

Step 3. Deriving numerous input conditions by a diversified search of subregions

In Step 3, MRS-PRIM (or DRS-PRIM) is fitted to the training set via multiple independent stochastic trials. Figure 1 visually compares the response optimization architecture in between deterministic and stochastic methods. Unlike deterministic architecture, stochastic architecture enables a total of $K \cdot T$ number of independent and diversified stochastic trials to be implemented. K different options ($S_1 \sim S_K$) are prepared for input feature subsample size S , and T independent stochastic trials are implemented for each of the K options. Note that $S_1 \sim S_K$ and T can be freely determined by the process engineer. A t th ($t : 1 \sim T$) individual stochastic trial under a k th ($k : 1 \sim K$) feature subsample size option S_k yields a unique box trajectory of its own. Upon its m th peeling step for determining B_m , $S(< P)$ number of input features is randomly subsampled from the input features $X = \{X_1, \dots, X_p, \dots, X_P\}$ as X^{subset} in Equation 10. Notably, subsampling (without replacement) of X^{subset} is established independently between every peeling step within each stochastic trial as well as between trials.

$$X^{subset} = \{X_1^{subset}, \dots, X_s^{subset}, \dots, X_S^{subset}\} \subset \{X_1, \dots, X_p, \dots, X_P\}. \quad (10)$$

By using the X^{subset} features sampled at the m th peeling step ($B_{m-1} \rightarrow B_m$), a set of $2S$ peeled candidate boxes C_{m-1} can be derived as in Equation (11). $C_{m-1}^{(s-)}$ and $C_{m-1}^{(s+)}$ denote the lower and upper percentile peeled boxes with respect to X_s^{subset} , which is an s th ($s : 1 \sim S$) sampled input feature among X^{subset} . Assuming that X_s^{subset} corresponds to X_p , the min-max subrange of $C_{m-1}^{(s-)}$ and $C_{m-1}^{(s+)}$ can be denoted as in Equations (12) and (13), respectively. Note that $[\Delta_{m-1,p}^{(-)}]$ and $[\Delta_{m-1,p}^{(+)}]$ are identical to Equation (8). The box among $2S$ candidate options in Equation (11) with the best box objective value is selected as the next box B_m . A series of consecutive peeling steps constitute a single t th ($t : 1 \sim T$) stochastic trial as a box peeling trajectory ($B_0 \rightarrow B_1 \rightarrow \dots \rightarrow B_m \rightarrow \dots$), which ends with respect to β_0 .

$$C_{m-1} = \left\{ C_{m-1}^{(1-)}, C_{m-1}^{(1+)}, \dots, C_{m-1}^{(s-)}, C_{m-1}^{(s+)}, \dots, C_{m-1}^{(S-)}, C_{m-1}^{(S+)} \right\}. \quad (11)$$

$$C_{m-1}^{(s-)} = [\Delta_{m-1,1}] \times \dots \times [\Delta_{m-1,p-1}] \times [\Delta_{m-1,p}^{(-)}] \times [\Delta_{m-1,p+1}] \times \dots \times [\Delta_{m-1,P}]. \quad (12)$$

$$C_{m-1}^{(s+)} = [\Delta_{m-1,1}] \times \dots \times [\Delta_{m-1,p-1}] \times [\Delta_{m-1,p}^{(+)}] \times [\Delta_{m-1,p+1}] \times \dots \times [\Delta_{m-1,P}]. \quad (13)$$

Suppose that the boxes constituting the peeling trajectory of a single trial are $B_0 \sim B_{m-1}$, with B_m having a lower support than the

β_0 restriction. The subregion boxes among $B_0 \sim B_{m-1}$ that do not pass the subregion validation are discarded. The post-validated box having the best box objective value within the peeling trajectory is selected as the optimal box for that single trial. MRS-PRIM defines its box objective as maximizing the $\bar{D}$ value of an identified subregion, assuming it has passed the validation on mean difference. Likewise, the box objective of DRS-PRIM is the MSE of an identified subregion, without being rejected by dual-validation on response mean (t -test) and dispersion (F -test). For both MRS-PRIM and DRS-PRIM, independently repeating an individual stochastic trial for T times is expected to yield T validated subregion boxes from T independent peeling paths. By implementing T stochastic trials across each of subspace sizes $S_1 \sim S_K$, a total of $K \cdot T$ decent boxes can be obtained as shown in Figure 1. Notably, setting $S = P$ results in an identical configuration to MR-PRIM or DR-PRIM, where stochasticity is absent within the peeling steps. In this case, repeating such a trial for T times yields T identical box trajectories, rendering multiple PRIM trials redundant.

Step 4. Selection of the optimal input condition for process control

Following the implementation of $K \cdot T$ stochastic trials in total, the process engineer finally selects a scope of distinct, validated subregion options among $K \cdot T$ boxes with best box objective values for process control. Notably, discovering numerous distinct subregions by a stochastic search of MRS-PRIM or DRS-PRIM makes the process control robust to disturbance factors that may incapacitate certain input range conditions. That is, even if some initially selected input conditions become infeasible for use, alternative options with similarly decent box objective values robustly remain for the process engineer to consistently select upon.

4 | Two Comparative Case Studies: Solving a Continuous Casting Stage MRO Problem and a Compressive Concrete Strength DRO Problem

In Section 4, two comprehensive case studies are conducted on MRO and DRO upon comparing optimization performance between stochastic and deterministic data-driven methods. In Section 4.1, an operational dataset collected from a continuous casting stage in a steel manufacturing process is optimized. The process stage objective to produce billets of desirable quality as a solidified form of molten iron [9]. Under N-fold cross validation, performance on MRO is compared in between MRS-PRIM, MR-PRIM, and MR-CART upon discovering validated input conditions with higher desirability. In Section 4.2, a compressive concrete dataset is dual optimized with respect to the concrete strength response variable. Performances of DRS-PRIM and DR-PRIM are compared under the MSE box objective via N-fold cross-validation. Section 4.3 establishes sensitivity analyses on MRO and DRO for validating the superior performance of stochastic methods over deterministic methods with respect to hyperparameter (minimal subregion size) control.

4.1 | Optimization of a Continuous Casting Stage via 4-Fold Cross Validation

Step 1. Preparing and preprocessing an operational dataset (for MRO)

The operational dataset of the continuous casting stage consists of $N = 5609$ observations, $J = 2$ response variables, and $P = 11$ input features. The two responses are return rate (Y_1 , %) and tensile strength (Y_2 , MPa), both of which are both LTB variables. The return rate refers to the ratio between the amount of iron scrap used in the process and the number of billets produced from it. Therefore, a higher Y_1 rate is always more desirable with respect to production cost. Similarly, an increase in the tensile strength of the produced billet is always assumed to be more desirable in quality. The target parameter values of Y_1 are set to $T_1^{lower} = 80$ and $T_1^{upper} = 100$. For Y_2 , the target values are predefined as $T_2^{lower} = 200$ and $T_2^{upper} = 300$. The shape parameters t_j are all set to one for simplicity. $Y_{i,1}$ and $Y_{i,2}$ are then transformed into $d_{i,1}$ and $d_{i,2}$ by desirability functions using the target values. An overall desirability function with equal weight parameters ($w_1 = w_2 = 0.5$) is applied, so that $D_i = \sqrt{d_{i,1} \cdot d_{i,2}}$.

Specifications of the 11 input features are as follows: elapsed time for a transfer of molten steel (X_1 , min), elapsed time for molten steel preparation (X_2 , min), amount of power consumption (X_3 , kW), amount of Dolomite stone input (X_4 , min), amount of Fe-Si input (X_5 , kg), amount of Si-Mn input (X_6 , kg), amount of Fe-Nb input (X_7 , kg), amount of recarburizer in the iron (X_8 , kg), moving time of a continuous casting machine (X_9 , min), waiting time of a continuous casting machine (X_{10} , min) and elapsed time of operation (X_{11} , min). Therefore, in terms of desirability, an i th observation can be denoted as $Obs_i(D_i, X_i) = \{D_i, X_{i,1}, X_{i,2}, \dots, X_{i,11}\}$.

To evaluate the generalized performance of the three MRO methods, the entire dataset is randomly split-folded into four subsets ($Subset_1$, $Subset_2$, $Subset_3$, $Subset_4$) containing 1402/1402/1402/1403 row observations each. Each of the four subsets is used as a testing set for each fold, and the remaining three subsets are used for fitting. For the first split-fold, for instance, the three MRO methods are, respectively, fitted onto the aggregated dataset of $Subset_2$, $Subset_3$, and $Subset_4$ so that the confirmation subregions can be found from $Subset_1$. Likewise, confirmation subregions of the second, third and fourth split-folds are found from $Subset_2$, $Subset_3$ and $Subset_4$, respectively. Throughout each split-fold, the performances of MRS-PRIM, MR-PRIM and MR-CART are compared with respect to the box objective $\bar{D}$ under the two-sample t -test.

Step 2. Defining α (peeling hyperparameter) and β_0 (stopping hyperparameter)

In Step 2, the values of hyperparameters α and β_0 are predefined for MRS-PRIM and MR-PRIM. Peeling hyperparameter α is set as 0.05, a typical value for patient peeling. By doing so, observations with the lowest (or highest) 5% percentiles of an input feature

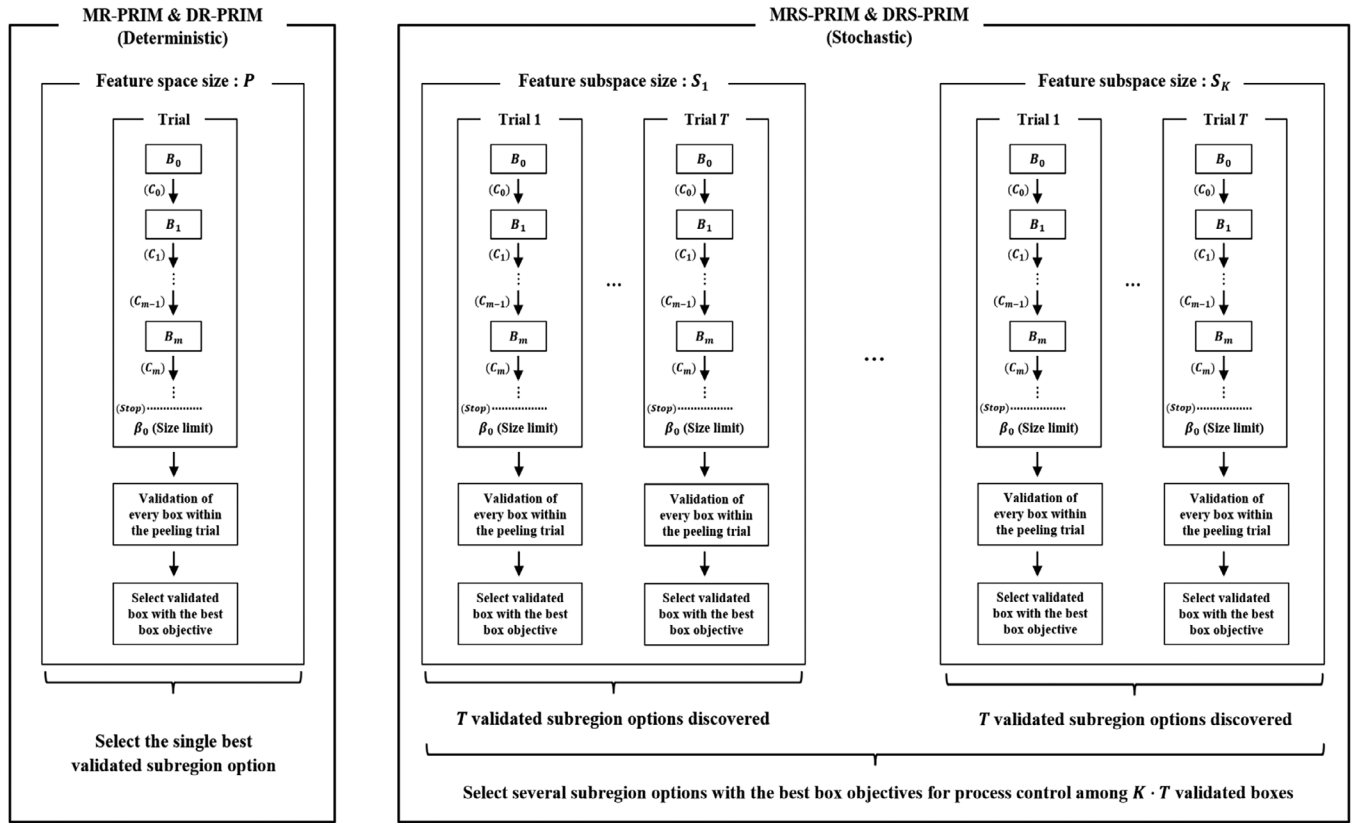


FIGURE 1 | Comparison of the deterministic (MR-PRIM, DR-PRIM) and stochastic (MRS-PRIM, DRS-PRIM) response optimization architecture.

are peeled away from the box in each peeling step. Minimal subregion size is fixed as 150, so that a box peeling trajectory ends when the observational size becomes smaller than 150. For a fair comparison with PRIM-based methods, the minimal leaf node size for MR-CART is also set as 150 so that the nodes with a smaller size than 150 are pruned. The hyperparameters are set as identical across four split-folds.

Step 3. Deriving numerous input conditions by a diversified search of subregions

In Step 3, the three MRO methods are fitted onto the data across each of four train-test split folds. For MRS-PRIM, $T = 500$ distinct stochastic trials are implemented with a feature subset scope of $K = 5$. Subset size S ranges from $6 \sim 10$ ($S_1 = 6, S_2 = 7, S_3 = 8, S_4 = 9, S_5 = 10$). Note that the decision of T and K depends on the discretion of a process engineer, but we assume that $K \cdot T = 2500$ stochastic trials are enough to yield numerous options for highly desirable input conditions. Within the box trajectory of an individual stochastic trial, the boxes that survive the t -test validation are preserved. Among the surviving boxes, the single identified subregion with the highest $\bar{D}$ is selected as the best subregion for that individual trial. Subregion validation is established using a two-sample t -test with a p value of 0.1, which is a widespread approach for validating subregion desirability on MRO [4, 9, 16]. Eventually, 2500 independent stochastic trials yield 2500 identified subregions that survived the t -test with decent desirability. For MR-CART, t -test validation is applied to

every terminal leaf node, ensuring that only the best identified subregions that have been t -test validated can be selected within its single decision tree. For MR-PRIM, the single identified subregion having the highest $\bar{D}$ among boxes that have passed the t -test validation is selected within its single PRIM trial.

Table 2 reports the most desirable subregions derived from each of the three MRO methods with the highest desirability across 4 split-folds. Top 10 subregions with the highest box objective at each fold are reported from MRS-PRIM and MR-CART, along with a single subregion derived from MR-PRIM. The best performing subregion at each split-fold is underlined. Sample mean values of response variables ($\bar{y}_1, \bar{y}_2$) are reported as well, where $\bar{y}_1$ closer to $T_1^{upper} (= 100)$ and $\bar{y}_2$ closer to $T_2^{upper} (= 300)$ are always more desirable. Throughout each of the four split folds, MRS-PRIM consistently outperforms the other two methods by finding multiple subregions with better box objective values. Such a robust performance on t -test validation originates from the algorithmic aspect of MRS-PRIM, employing numerous stochastic trials to eventually increase the likelihood of discovering various desirable input conditions among which multiple options of validated subregions robustly remain.

Step 4. Selection of the optimal input condition for process control

In Step 4, an optimal input condition is selected by the process engineer for actual process control. Supposing that the process

TABLE 2 | Comparison of the validated subregions in between data-driven MRO methods across four split-folds (minimal size of 150).

Split-fold	Fold 1				Fold 2				Fold 3				Fold 4			
MRO method	Box objective (p value)	Sample response mean: $\bar{y}_1/\bar{y}_2$	S size at trial	Box objective (p value)	Sample response mean: $\bar{y}_1/\bar{y}_2$	S size at trial	Box objective (p value)	Sample response mean: $\bar{y}_1/\bar{y}_2$	S size at trial	Box objective (p value)	Sample response mean: $\bar{y}_1/\bar{y}_2$	S size at trial	Box objective (p value)	Sample response mean: $\bar{y}_1/\bar{y}_2$	S size at trial	
MRS-PRIM (Stochastic)	0.559 (0.32)	93.0/254.8	8	0.567 (0.31)	93.0/256.1	6	0.569 (0.12)	92.9/257.0	6	0.585 (0.16)	92.9/260.4	8	0.585 (0.16)	92.9/260.4	8	
	0.559 (0.88)	92.6/256.1	9	0.567 (0.11)	92.6/258.0	9	0.569 (0.11)	92.8/256.7	6	0.583 (0.32)	93.1/259.2	7	0.583 (0.32)	93.1/259.2	7	
	0.557 (0.98)	92.8/255.2	6	0.565 (0.11)	92.8/256.6	7	0.569 (0.11)	92.6/257.6	6	0.580 (0.17)	92.9/258.9	6	0.580 (0.17)	92.9/258.9	6	
	0.556 (0.82)	92.8/255.2	6	0.565 (0.19)	93.0/256.6	9	0.568 (0.14)	92.8/256.4	8	0.577 (0.18)	93.0/258.0	6	0.577 (0.18)	93.0/258.0	6	
	0.556 (0.92)	92.7/256.1	7	0.563 (0.14)	92.6/257.1	7	0.566 (0.16)	92.9/255.9	8	0.577 (0.12)	92.8/258.9	6	0.577 (0.12)	92.8/258.9	6	
	0.556 (0.82)	92.8/255.2	7	0.563 (0.28)	92.9/256.3	9	0.566 (0.15)	92.6/257.1	8	0.577 (0.15)	93.2/257.7	9	0.577 (0.15)	93.2/257.7	9	
	0.554 (0.81)	92.8/254.9	7	0.558 (0.11)	92.8/255.8	7	0.565 (0.12)	92.6/256.9	6	0.575 (0.23)	92.7/258.4	6	0.575 (0.23)	92.7/258.4	6	
	0.553 (0.68)	93.0/254.0	7	0.557 (0.31)	92.8/255.4	6	0.565 (0.15)	92.8/256.2	7	0.574 (0.33)	92.5/259.8	6	0.574 (0.33)	92.5/259.8	6	
	0.553 (0.75)	92.6/255.6	6	0.556 (0.22)	92.3/257.3	6	0.562 (0.14)	92.8/256.1	7	0.574 (0.13)	92.3/260.5	8	0.574 (0.13)	92.3/260.5	8	
MR-CART (Deterministic)	0.553 (0.81)	92.4/256.0	7	0.555 (0.81)	92.9/255.0	6	0.558 (0.10)	92.7/255.5	6	0.572 (0.61)	92.2/260.2	6	0.572 (0.61)	92.2/260.2	6	
	0.538 (0.36)	92.6/254.4		0.538 (0.12)	92.5/254.7		0.546 (0.26)	92.6/254.2		0.526 (0.75)	92.3/252.7		0.526 (0.75)	92.3/252.7		
	0.525 (0.76)	92.6/250.5		0.536 (0.31)	92.4/252.8		0.544 (0.13)	92.2/255.8		0.525 (0.93)	92.6/251.2		0.525 (0.93)	92.6/251.2		
	0.523 (0.90)	92.2/251.7		0.516 (0.93)	92.6/249.4		0.537 (0.22)	92.2/253.5		0.506 (0.55)	92.2/250.1		0.506 (0.55)	92.2/250.1		
	0.506 (0.46)	92.2/249.2		0.505 (0.22)	92.2/250.0		0.509 (0.93)	92.4/250.5		0.480 (0.28)	90.0/253.6		0.480 (0.28)	90.0/253.6		
	0.499 (0.25)	91.6/249.7		0.492 (0.53)	91.3/251.0		0.493 (0.22)	91.9/249.7		0.480 (0.88)	92.0/246.8		0.480 (0.88)	92.0/246.8		
	0.489 (0.77)	90.2/255.5	11 (All)	0.485 (0.13)	90.9/250.5	11 (All)	0.487 (0.25)	91.5/249.1	11 (All)	0.466 (0.45)	90.8/250.6	11 (All)	0.466 (0.45)	90.8/250.6	11 (All)	
	0.459 (0.11)	90.1/249.7		0.484 (0.60)	90.8/251.3		0.486 (0.51)	91.8/249.0		0.458 (0.12)	90.8/248.2		0.458 (0.12)	90.8/248.2		
	0.450 (0.21)	90.2/248.2		0.465 (0.80)	90.6/249.7		0.485 (0.26)	90.0/255.0		0.441 (0.97)	89.6/249.5		0.441 (0.97)	89.6/249.5		
MR-PRIM (Deterministic)	0.447 (0.66)	89.7/250.1		0.465 (0.15)	90.2/250.9		0.476 (0.77)	91.2/247.8		0.441 (0.10)	89.3/250.0		0.441 (0.10)	89.3/250.0		
	0.435 (0.34)	89.6/250.3		0.464 (0.65)	91.3/246.2		0.470 (0.77)	91.4/248.4		0.409 (0.36)	88.5/247.3		0.409 (0.36)	88.5/247.3		
	0.534 (0.47)	92.2/254.1		0.516 (0.30)	91.8/253.4		0.512 (0.14)	92.0/251.8		0.518 (0.26)	91.7/253.6		0.518 (0.26)	91.7/253.6		

engineer has established a particular split-fold among four, any subregion from that split-fold can be selected as an activation option. Table 3 specifies the information of the 10 most desirable input conditions of MRS-PRIM under the fourth split-fold, as well as the single best subregion from MR-CART and MR-PRIM. Specifications include input subrange information of 11 features, and the predefined subset size S of the trial from which the subregion was found. Suppose that a process engineer has selected subregion 1 (first row) in Table 3 with the highest box objective. Then, the amount of time required for transferring molten steel (X_1) is controlled in between 13 and 40 min. A min-max range of time allowed for molten steel preparation (X_2) is 17–36 min, and the amount of power consumption (X_3) is 2300–3700 kW. The amount of Dolomite stone (X_4) is controlled at 200–295 kg, and Fe-Si (X_5) is not utilized. The amounts of Si-Mn input (X_6), Fe-Nb input (X_7), and recarburizer input (X_8) are controlled within the range of 180–770, 7–13, and 30–210 kg, respectively. Moving time (X_9) and waiting time (X_{10}) are adjusted within 1–24 and 9–34 min, respectively. Finally, the time of operation (X_{11}) must be controlled to 40–42 min. By adjusting the input subrange as such for process control, the process engineer can expect a desirable extent of response values for both return rate (Y_1) and tensile strength (Y_2).

Assuming that the process engineer has established a fourth split-fold, the distinct subregion options of MRS-PRIM reported in Table 3 differ in at least one or more of the 11 input subranges ($\Delta_{X_1} \sim \Delta_{X_{11}}$). Such a diversity upon process control options implies robustness on online disturbance factors that impose limits on controllability of certain input features and their subranges. For instance, suppose that a process engineer has encountered the following real-time restrictions at the continuous casting process: i) power consumption (X_3) must remain below (or same as) 4000, ii) usage of Fe-Si (X_5) is required above zero, iii) Fe-Nb input range (X_7) must include zero, and iv) moving time (X_9) should not exceed 23 min at any chance. Among the 10 best options of MRS-PRIM in Table 3, the options 3, 5, 7, 9, and 10 fully meet the requirements above. Meanwhile, the best subregion of MR-CART (and MR-PRIM) fails to do so. Thus, it can be inferred that preparing diversified subrange options in advance helps the process engineer establish a robust process control to real-time changing restrictions upon the manufacturing environment, while maintaining the box objective value within a decent scope.

4.2 | Optimization of a Compressive Concrete Dataset via 3-Fold Cross Validation

Step 1. Preparing and preprocessing an operational dataset (for DRO)

The compressive concrete dataset consists of $N = 1030$ observation rows, with compressive strength value of a concrete as its univariate response variable Y and $J = 8$ input features ($X_1 \sim X_8$). An i th realization is denoted as $Obs_i(y_i, x_i) = \{y_i, x_{i,1}, x_{i,2}, \dots, x_{i,8}\}$. The target constant (*Target*) is predefined as 60, implying that an optimal subregion contains realized Y_i values densely concentrated around 60. The input feature specifications are as follows: cement input (X_1), blast furnace slag input (X_2), fly ash input (X_3), water input (X_4), superplasticizer input (X_5),

TABLE 3 | Input subrange specifications of the best validated subregions derived by MRO methods (at fold 4, minimal size of 150).

MRO method	Subregion rank (Within-method)	Box objective: Validated $\bar{D}$	Min-max subrange information of controllable input features ($X_1 \sim X_{11}$)											S size at trial
			Δ_{X_1}	Δ_{X_2}	Δ_{X_3}	Δ_{X_4}	Δ_{X_5}	Δ_{X_6}	Δ_{X_7}	Δ_{X_8}	Δ_{X_9}	$\Delta_{X_{10}}$	$\Delta_{X_{11}}$	
MRS-PRIM	1	0.585	[13,40]	[17,36]	[2300,3700]	[200,295]	[0,0]	[180,770]	[7,13]	[30,210]	[1,24]	[9,34]	[40,42]	8
	2	0.583	[13,45]	[24,36]	[2800,3700]	[200,295]	[0,0]	[180,770]	[3,13]	[30,210]	[1,13]	[12,35]	[40,44]	7
	3	0.580	[13,28]	[15,31]	[3000,4000]	[200,280]	[0,75]	[0,740]	[0,9]	[70,150]	[1,12]	[8,34]	[39,163]	6
	4	0.577	[13,47]	[15,30]	[3000,4100]	[200,280]	[0,75]	[0,740]	[0,9]	[70,210]	[1,12]	[8,36]	[39,163]	6
	5	0.577	[13,29]	[15,30]	[3000,3900]	[200,280]	[0,75]	[0,740]	[0,9]	[70,210]	[1,20]	[8,37]	[39,163]	6
	6	0.577	[13,109]	[14,40]	[2300,4300]	[200,295]	[0,0]	[130,865]	[8,13]	[20,195]	[4,43]	[9,33]	[40,44]	9
	7	0.575	[13,28]	[15,31]	[3000,3300]	[200,200]	[0,75]	[0,1000]	[0,12]	[30,180]	[0,13]	[8,39]	[39,163]	6
	8	0.574	[13,40]	[17,32]	[2700,3700]	[200,205]	[0,0]	[180,720]	[5,12]	[30,210]	[3,24]	[10,35]	[39,42]	6
	9	0.574	[13,40]	[15,28]	[3000,3300]	[200,285]	[0,75]	[0,1000]	[0,13]	[30,180]	[0,13]	[10,40]	[39,163]	8
	10	0.572	[13,29]	[15,31]	[3000,3400]	[200,285]	[0,75]	[0,740]	[0,9]	[30,180]	[1,12]	[8,43]	[39,163]	6
MR-CART	1	0.526	[10,90]	[15,32]	[2000,4500]	[0,400]	[0,0]	[0,495]	[0,8]	[60,180]	[0,25]	[2,55]	[39,44]	11
MR-PRIM	1	0.518	[13,93]	[15,37]	[3000,4000]	[100,285]	[0,80]	[0,1000]	[0,20]	[20,240]	[0,30]	[8,37]	[39,163]	(All)

coarse aggregate input (X_6), fine aggregate input (X_7), and age by day (X_8). Considering the observational size of the dataset, 3-fold cross-validation is applied for evaluating the optimization performance of DRS-PRIM and DR-PRIM. The rows are randomly split into three subsets ($Subset_1$, $Subset_2$, $Subset_3$) with 343/343/344 observations each. DRS-PRIM and DR-PRIM are fitted onto the aggregated subset $Subset_2$ and $Subset_3$, and an identified subregion is then validated using the confirmation subregion within $Subset_1$. The confirmation subregions for the second and third split-fold are found from $Subset_2$ and $Subset_3$, respectively.

Step 2. Defining α (peeling hyperparameter) and β_0 (stopping hyperparameter)

We set $\alpha = 0.05$, which is a typical peeling ratio for PRIM-based methods in data-driven DRO. The minimum available size for a subregion box of DRS-PRIM and DR-PRIM is fixed as 130 across three split-folds for a fair comparison.

Step 3. Deriving numerous input conditions by a diversified search of subregions

In Step 3, DRS-PRIM and DR-PRIM are, respectively, fitted across three split-folds for discovering identified subregions. For DRS-PRIM, the number of trials T is set as 300 and the scope of subset size S as $K = 3$ ($S_1 = 5$, $S_2 = 6$, $S_3 = 7$). Therefore, $K \cdot T = 900$ independent stochastic trials are implemented at each split-fold. Upon each stochastic trial, the box with a best box objective (lowest identified subregion MSE) is selected among the dual-validated boxes that constitute the trial. The dual-validation for a subregion is conducted on every box within each trial, by applying the F -test and t -test. A subregion box is validated if and only if not rejected by both tests, implying that both location and scale do not significantly differ between the identified and confirmation subregion. A total of 900 dual-validated subregion boxes can then be aggregated across 900 independent trials, and the best boxes among them can be selected as final options by the process engineer. For DR-PRIM, the single dual-validated box with a lowest box objective is selected as the optimal subregion.

Table 4 compares the derived subregions of DRS-PRIM and DR-PRIM across three split-folds. The MSE values for identified subregion and confirmation subregion are reported, along with scope size S and p values of t -test and F -test. We suppose that among 900 dual-validated subregions of DRS-PRIM, the process engineer prefers to extract top 30 options with the lowest box objective (identified subregion MSE value) at first, and then select the 10 among them having the lowest confirmation subregion MSE s as the final scope of options. Table 4 specifies the performance of those 10 MRS-PRIM subregions, sorted by the box objective. The option having the lowest box objective is underlined at each split-fold. Notably, DRS-PRIM consistently discovers multiple subregion options with better box objectives compared to DR-PRIM. The subregions of DRS-PRIM come from different backgrounds of S sizes, implying that the accumulation of numerous independent PRIM trials via random subspace

empowers the algorithmic search for dual-optimal subregions throughout all split-folds.

Step 4. Selection of the optimal input condition for process control

In Step 4, optimal input condition is selected among the results derived by DRS-PRIM. Table 4 has shown that regardless of the split-fold, DRS-PRIM robustly derives several distinct input conditions with better box objective values than DR-PRIM. Table 5 specifies the min-max input subranges of the 10 dual-validated subregions of DRS-PRIM and a single subregion of DR-PRIM reported in Table 4. Notably, the min-max subrange information of 10 distinct subregions of DRS-PRIM mostly differs across input features $X_1 \sim X_5$, while exhibiting high similarity upon X_6 (811 $\sim$ 1134.3), X_7 (605 $\sim$ 992.6), and X_8 (7 $\sim$ 100). This implies that close-to-target ($Target = 60$) observations are mostly concentrated within the aforementioned $X_6 \sim X_8$ local subrange, and distinct subregions within that subrange can be further verified by varying the subranges of $X_1 \sim X_5$. Therefore, the discretion of the process engineers to flexibly select upon subrange options for process control is allowed with respect to $X_1 \sim X_5$, but is relatively limited upon $X_6 \sim X_8$.

Suppose that the process engineer selects the very best subrange option of DRS-PRIM (underlined in Table 4). The cement input (X_1) is then controlled within the range of 182–475, blast furnace slag (X_2) in between 15 and 282.8, fly ash in between 0 and 163, water input (X_4) within 126.6 and 192.5, superplasticizer (X_5) in between 5.5 and 32.2, coarse aggregate (X_6) in between 811 and 1134.3, fine aggregate (X_7) within 605–992.6 and day length (X_8) in between 7 and 100. Similar to the conclusions of MRO in Section 4.1, DRS-PRIM consistently outperforms DR-PRIM by incorporating the random subspace technique with hundreds of PRIM trials to intensively search for input conditions that dual-optimize the concrete strength.

4.3 | Sensitivity Analysis: Minimum Subregion Size Control

Section 4.3 presents a sensitivity analysis on comparing the performance of stochastic methods (MRS-PRIM, DRS-PRIM) with deterministic methods (MR-CART, MR-PRIM, DR-PRIM) with respect to minimum subregion size control. The scope of analysis in Sections 4.1 and 4.2 is thus extended in Section 4.3 by introducing the minimum subregion size (or β_0) as a controllable factor for evaluating the generalized performance of each method. Reducing the minimum subregion size enables additional peeling or partitioning, thereby creating opportunities to discover new, narrower subregions with better box objectives. However, if a method remains vulnerable to subregion validation, additionally discovered subregions with better box objectives may be discarded due to rejection upon statistical tests, and the method may eventually fail upon yielding a consistent improvement in its box objective. Accordingly, the aim of sensitivity analysis is to verify whether MRS-PRIM and DRS-PRIM can consistently discover numerous subregions with better box objective values than deterministic methods throughout

TABLE 4 | Comparison of the dual-validated subregions in between data-driven DRO methods across 3 split-folds (minimal size of 130).

Split-fold	Fold 1					Fold 2					Fold 3				
	Box objective: Identified subregion MSE ($\bar{Y}/\hat{\delta}$)	Confirmation subregion MSE ($\bar{Y}/\hat{\delta}$)	S size at trial	p values (t-test / F-test)	Box objective: Identified subregion MSE ($\bar{Y}/\hat{\delta}$)	Confirmation subregion MSE ($\bar{Y}/\hat{\delta}$)	S size at trial	p values (t-test / F-test)	Box objective: Identified subregion MSE ($\bar{Y}/\hat{\delta}$)	Confirmation subregion MSE ($\bar{Y}/\hat{\delta}$)	S size at trial	p values (t-test / F-test)			
DRO method	189.18 (55.83/13.10)	203.04 (54.15/12.99)	5	0.39/0.95	174.74 (55.41/12.39)	221.13 (56.37/14.42)	5	0.66/0.17	209.95 (54.51/13.41)	211.64 (54.97/13.65)	5	0.82/0.85			
	197.54 (55.49/13.31)	194.77 (54.22/12.70)	5	0.52/0.68	179.93 (55.39/12.59)	223.52 (55.21/14.16)	5	0.93/0.27	211.41 (54.54/13.47)	212.44 (55.03/13.70)	5	0.81/0.86			
	198.66 (55.56/13.37)	218.36 (53.61/13.32)	5	0.33/0.99	182.88 (55.49/12.75)	220.75 (55.66/14.21)	5	0.94/0.31	213.53 (54.31/13.45)	210.62 (54.82/13.55)	5	0.80/0.93			
	199.77 (55.09/13.25)	223.06 (53.19/13.29)	5	0.33/0.96	182.91 (55.17/12.63)	228.85 (55.06/12.30)	5	0.96/0.25	213.92 (54.19/13.42)	210.06 (54.91/13.57)	5	0.72/0.90			
	202.28 (55.35/13.44)	201.24 (54.07/12.89)	5	0.52/0.71	183.36 (55.16/12.64)	212.24 (55.70/13.92)	5	0.80/0.37	214.98 (54.38/13.54)	211.17 (55.13/13.69)	6	0.71/0.90			
	203.40 (55.10/13.39)	205.29 (53.84/12.93)	6	0.52/0.76	184.37 (55.44/12.79)	216.09 (55.40/13.96)	5	0.98/0.41	216.53 (54.20/13.52)	207.91 (54.84/13.46)	5	0.74/0.98			
	203.59 (54.83/13.30)	221.14 (53.16/13.20)	6	0.39/0.96	185.13 (55.48/12.83)	207.94 (56.02/13.86)	6	0.80/0.47	217.89 (54.33/13.63)	184.96 (55.05/12.64)	5	0.71/0.51			
	204.21 (55.41/13.53)	182.83 (55.23/12.65)	6	0.93/0.56	186.84 (55.33/12.84)	228.82 (55.30/14.37)	5	0.99/0.29	218.75 (54.12/13.57)	211.08 (54.28/13.35)	6	0.94/0.90			
	206.42 (54.96/13.45)	201.57 (54.32/13.01)	6	0.74/0.77	188.09 (55.16/12.83)	212.98 (55.67/13.94)	5	0.81/0.44	218.85 (53.99/13.52)	207.82 (55.02/13.53)	5	0.61/0.98			
	208.00 (55.14/13.58)	180.95 (55.04/12.50)	5	0.96/0.46	188.46 (54.89/12.74)	230.14 (54.95/14.30)	5	0.98/0.27	219.82 (54.09/13.60)	187.51 (55.03/12.76)	5	0.63/0.57			
DR-PRIM	208.13 (54.75/13.44)	249.38 (52.45/13.87)	8 (All)	0.25/0.74	199.38 (55.12/13.25)	223.41 (55.57/14.27)	8 (All)	0.84/0.48	248.93 (53.60/14.42)	248.05 (53.60/14.39)	8 (All)	0.99/0.99			
(Deterministic)															

TABLE 5 | Input subrange specifications of the best dual-validated subregions derived by DRO methods (at fold 3, minimal size of 130).

DRO method	Subregion rank (Within-method)	Box Objective: Dual-validated		Min-max subrange information of controllable input features ($X_1 \sim X_8$)								S size at trial
		MSE	Δ_{X_1}	Δ_{X_2}	Δ_{X_3}	Δ_{X_4}	Δ_{X_5}	Δ_{X_6}	Δ_{X_7}	Δ_{X_8}		
DRS-PRIM	1	209.95	[182, 475]	[15, 282.8]	[0, 163]	[126.6, 195.5]	[5.5, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		5
	2	211.41	[178, 475]	[15, 282.8]	[0, 141]	[126.6, 195.5]	[3.6, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		5
	3	213.53	[173.8, 475]	[17.2, 282.8]	[0, 163]	[126.6, 201.3]	[5.9, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		5
	4	213.92	[178, 475]	[15, 282.8]	[0, 163]	[126.6, 195]	[3.6, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		5
	5	214.98	[182, 475]	[15, 282.8]	[0, 141]	[126.6, 195.4]	[4.6, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		6
	6	216.53	[173.5, 475]	[17.2, 282.8]	[0, 173.5]	[126.6, 200.6]	[5.9, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		5
	7	217.89	[165, 475]	[17.2, 282.8]	[0, 163]	[126.6, 194.9]	[7, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		5
	8	218.75	[165, 475]	[17.2, 282.8]	[0, 163]	[126.6, 186]	[6.7, 32.2]	[822, 1134.3]	[605, 992.6]	[7, 100]		6
	9	218.85	[173.5, 475]	[20, 282.8]	[0, 173.5]	[126.6, 195]	[3.6, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		5
	10	219.82	[165, 475]	[17.2, 282.8]	[0, 167]	[126.6, 194.9]	[7, 32.2]	[811, 1134.3]	[605, 992.6]	[7, 100]		5
DR-PRIM	1	248.93	[164, 491]	[15, 282.8]	[0, 163.8]	[121.8, 246.9]	[3.9, 32.2]	[811, 1134.3]	[605, 992.6]	[14, 100]		8 (All)

differing β_0 settings, while staying robust to subregion validation test.

Figure 2 visualizes the box objective ranges of the 10 best validated subregions derived by MRS-PRIM (blue vertical bar) and MR-CART (brown vertical bar) across four split-folds and six different size limits. The blue and brown dots located at the middle of the bars denote the average of validated $\bar{D}$ values among the 10 best subregions of each method, and the green dot denotes the validated $\bar{D}$ value of a single-derived subregion of MR-PRIM. As in Section 4.1, hyperparameters are set as $T = 500$ and $K = 5$. Two-sample t -test is conducted for subregion validation under a p value of 0.1. Note that the results reported under a minimum size of 150 in Figure 2 correspond to those of Section 4.1.

The blue dots of MRS-PRIM consistently remain above the brown and green dots throughout split-folds, implying that it outperforms deterministic MRO methods by robustly discovering multiple validated subregions with better desirability. MRS-PRIM conducts abundant stochastic PRIM trials to magnify the likeliness of identifying improved subregions along with a decrease in β_0 , leveraging the non-decreasing nature of each PRIM trajectory. With respect to a particular trial of MRS-PRIM, only the boxes passing the t -test are retained, and the box with the highest $\bar{D}$ among them is selected as the optimal subregion for that single peeling trial (or trajectory). Once a trajectory has been established under a predefined β_0 , extending the trajectory by reducing the β_0 does not alter the previously accumulated peeling record, unless discarding the trajectory. Therefore, the box objective within a trajectory increases if and only if an additional peeling reveals a better subregion that also passes a t -test. Notably, the exact values of the blue dots of MRS-PRIM in Figure 2 show a consistent increase along with β_0 reduction, which are as follows: first fold ($0.554 \rightarrow 0.559 \rightarrow 0.571 \rightarrow 0.580 \rightarrow 0.604 \rightarrow 0.646$), second fold ($0.562 \rightarrow 0.565 \rightarrow 0.568 \rightarrow 0.571 \rightarrow 0.606 \rightarrow 0.668$), third fold ($0.566 \rightarrow 0.569 \rightarrow 0.579 \rightarrow 0.589 \rightarrow 0.617 \rightarrow 0.654$), and fourth fold ($0.577 \rightarrow 0.586 \rightarrow 0.601 \rightarrow 0.615 \rightarrow 0.628 \rightarrow 0.685$).

Meanwhile, MR-PRIM remains vulnerable to subregion validation if its only peeling path suffers from the statistical test. For instance, the green dot value of MR-PRIM within the third split-fold shows no improvement along a decrease in β_0 . The green dot denotes a box objective value of 0.512, having a smallest subregion size (same as or larger than size 25) that passes the statistical test within the single trajectory of MR-PRIM. Such instance shows that a non-decreasing nature of an embedded PRIM is not equivalent to yielding a consistent increase in the box objective along with a decline in β_0 . For a consistent discovery of newer and better subregions while remaining robust to validation, an intensive search of subregions must be entailed such as by MRS-PRIM. Employing a randomized scope of input features upon the local peeling decisions has been integrated with an enormous number of independent PRIM-based trials, eventually amplifying the performance of MRS-PRIM.

The architecture of MR-CART consists only of a single CART tree, in which the less desirable terminal nodes inevitably coexist with desirable nodes. Figure 2 shows that the lower tips of the brown bar (10th best subregion $\bar{D}$) often perform below 0.5. Yet, it is notable that upon certain minimal subregion sizes (e.g., minimal sizes 50 and 75 in third split-fold, minimal size 75 in fourth split-

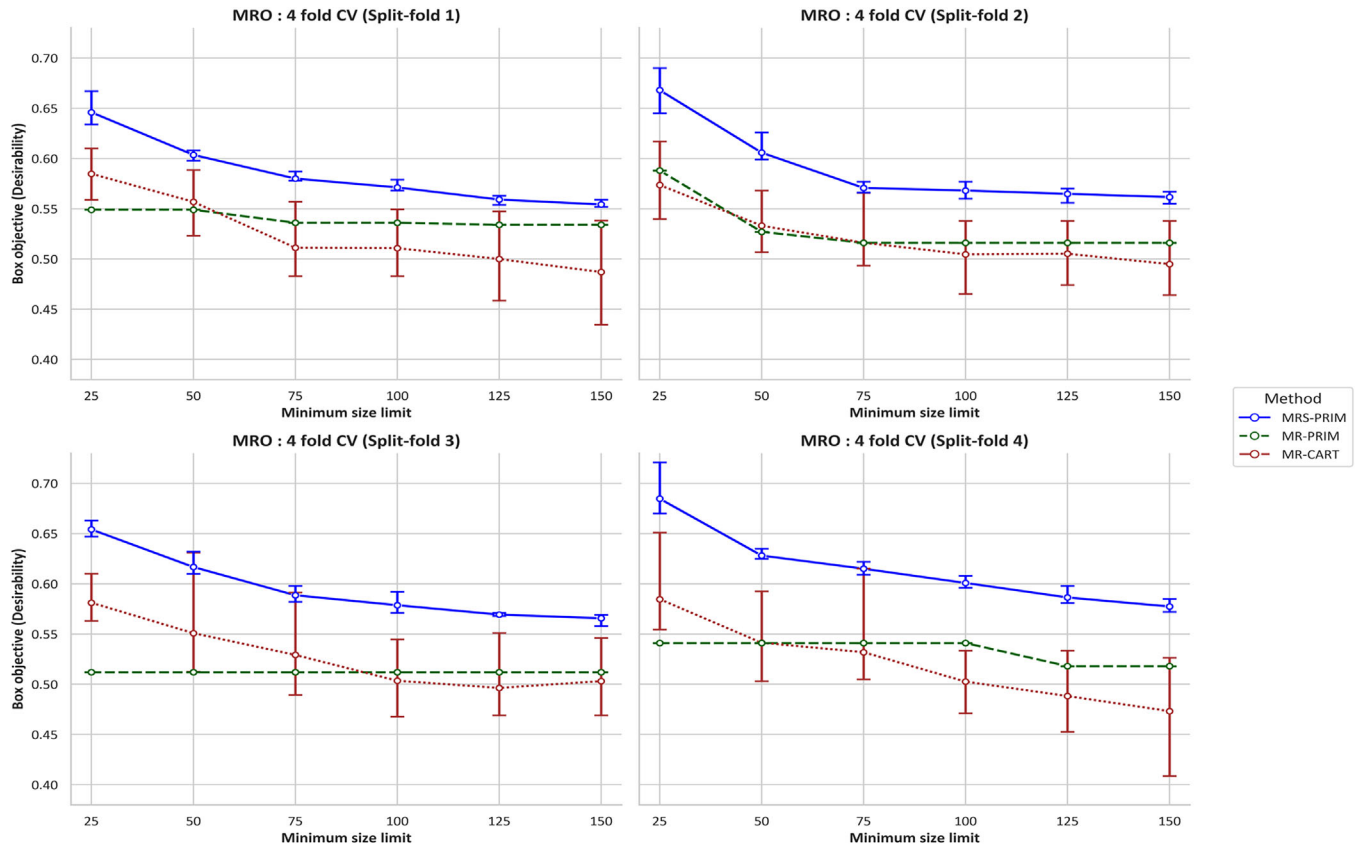


FIGURE 2 | Performance comparison of stochastic and deterministic MRO methods under varying minimum subregion size limits (4-fold CV).

fold), the upper tip of the brown bar trespasses the blue bar. Such instances show that multiple terminal nodes of MR-CART have the potential to produce highly desirable subregions, the best of which may perform almost as decent as MRS-PRIM. This is because as β_0 decreases, MR-CART is likely to increase its depth and produce a larger number of terminal nodes owing to its binary partitioning. Meanwhile, a decline in the upper tip value of the brown bar of MR-CART can be discovered at certain intervals (i.e., $50 \rightarrow 25$ at third split-fold, $75 \rightarrow 50$ at fourth split-fold), whereas the upper tip of the blue bar improves upon the same intervals. Such a contrast occurs because although a decrease in minimal subregion size brought about an additional chance for MR-CART to increase its tree depth, the newly discovered terminal nodes (or identified subregions) with better desirability could not survive the t -test validation.

Figure 3 compares the optimal MSE values of the dual-validated subregions derived by DRS-PRIM and DR-PRIM throughout three split-folds and five subregion size limits (30/55/80/105/130). For a fair comparison, $T = 300$ and $K = 3$ are fixed as in Section 4.2. It is supposed that 30 dual-validated subregions of DRS-PRIM with the smallest box objective (identified subregion MSE) are selected at first, and the 10 among them with the smallest confirmation subregion MSE are finally selected by the process engineer. Thus, the results reported under a minimum size of 130 in Figure 3 are identical to the derivations of Section 4.2. The MSE values of those 10 best subregions of DRS-PRIM are denoted in round black dots, and those of DR-PRIM in green (or red) triangles. The red triangles (denoted as “DR-PRIM (N)”) imply that although β_0 had been updated by a single-step decline for

an additional peeling opportunity, DR-PRIM could not discover a better subregion that passes dual-validation.

Implications of Figure 3 are summarized as follows: firstly, at least 10 distinct DRS-PRIM options having a better box objective (identified subregion MSE) than DR-PRIM are found throughout all split-folds and β_0 settings. Secondly, DRS-PRIM consistently reports multiple black lines throughout β_0 settings that remain below the green (or red) line of DR-PRIM without line intersection, implying that DRS-PRIM outperforms DR-PRIM on both identified/confirmation MSE s. Although confirmation MSE is a supplementary metric utilized only for validation, its values tend to improve together with the identified MSE values upon DRS-PRIM. This is because DRS-PRIM already discovers numerous distinct subregions prior to dual-validation, increasing the likelihood for many of them to pass the tests so that the line-connected values likely improve together with a β_0 decline.

In accordance with a decline in β_0 , the 10 black lines of DRS-PRIM show a decreasing trend in general across split-folds, while consistently discovering new subregion options. In contrast, upon the β_0 intervals of $[105 \rightarrow 80]$ and $[80 \rightarrow 55]$ (at fold 1 and 3), and intervals of $[130 \rightarrow 105]$ and $[105 \rightarrow 80]$ and $[80 \rightarrow 55]$ (at fold 2), the red line of DR-PRIM remains unchanged. Such a non-improving performance of DR-PRIM stems from two possible scenarios: firstly, even if DR-PRIM had additionally peeled its box along with a single-step decline in β_0 , the narrowed subregion box has failed upon dual-validation. Secondly, the next optimal peeling step was expected to yield a box support lower than the upcoming smaller β_0 , so that additional decline

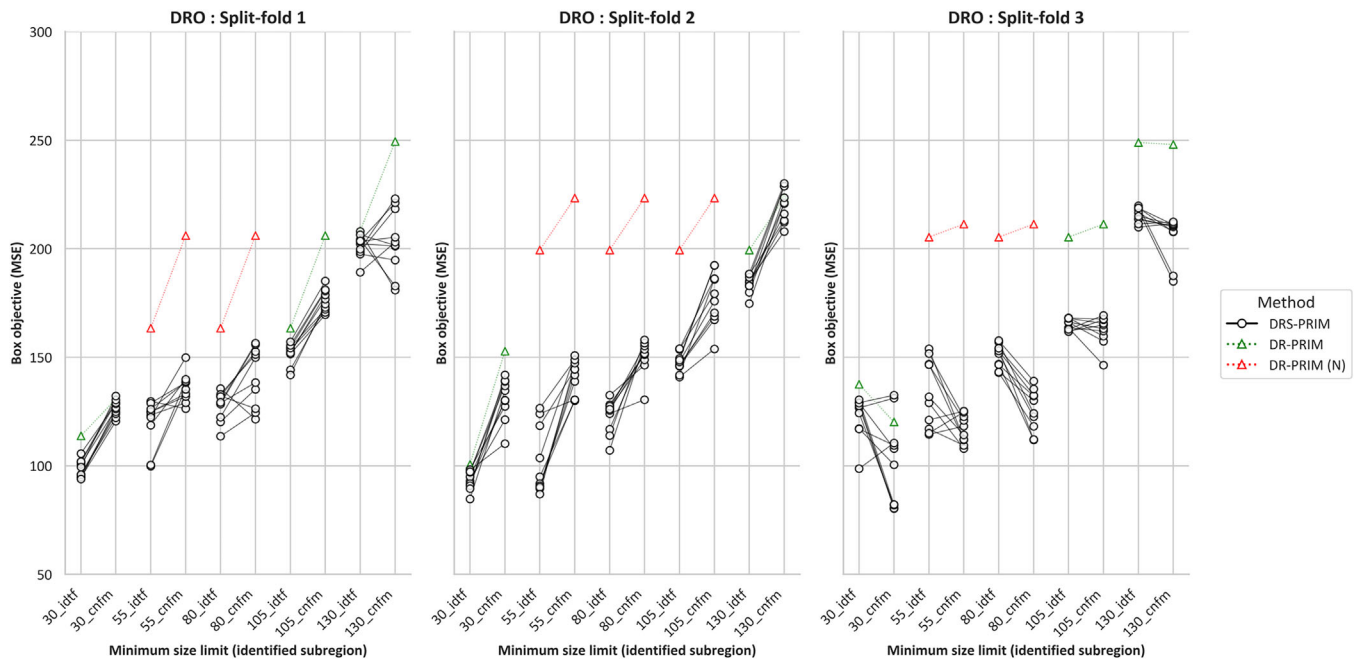


FIGURE 3 | Performance comparison of stochastic and deterministic DRO methods under varying minimum subregion size limits (3-fold CV).

in β_0 was required for the following peeling step to take place. The stochasticity of DRS-PRIM acts as a minimization strategy of those two risks, by computationally diversifying the peeling routes and aggregating those results to try to exploit every possible margin for improvement with respect to a β_0 decline.

5 | Concluding Remarks

Two new stochastic optimization methods have been proposed for deriving several gratifying input conditions from operational data. Their outstanding performance originates from the following strategies: (i) a proper design on objective functions, (ii) stochastic selection on the scope of features upon every peeling, and (iii) implementing numerous independent PRIM-based trials for a diversified and intensive search of optimal subregions. Two comprehensive case studies demonstrated that stochastic methods consistently discover better validated options than deterministic methods.

As a meaningful extension of our research, we propose developing a new stochastic method that can dual-optimize multiple response variables. A notable literature is Yang and Lee [10], extending a univariate DRO to multivariate responses by proper box objective designs. Yet, their work is inherently deterministic as of MR-PRIM or DR-PRIM in that a further diversification of peeling paths by an intensive search is infeasible. Therefore, developing new stochastic methods where multiple response variables are simultaneously and dually optimized will be a desirable extension of our work.

Another future research issue would be to develop a stochastic PRIM-based method for optimizing operational data collected from multiple process stages. Notable literature on applying PRIM for multistage optimization includes Kwak et al. [19] and Lee et al. [16]. Yet, they apply deterministic PRIM on multiple

process stages. By applying stochastic PRIM to derive optimal input conditions throughout stages, a new data-driven method for multistage optimization can be expected. Moreover, although our proposed methods effectively optimize operational data with a decently large number of rows and columns, situations in which the data size is extraordinarily large (e.g., millions of rows, hundreds of columns) have not been addressed. Accordingly, proposal of a scalable stochastic PRIM would further contribute to optimizing processes with gigantic data sizes. This is also left as a future work upon the field of data-driven MRO and DRO.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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